

Quiz 6

Instructions: You must show all of your work to receive credits for correct answers.

Problem 1. (6 points) Determine whether the sequence $\{a_n\}_{n=1}^{+\infty}$ converges or diverges and find the limit if it converges.

(a) $a_n = \frac{2-3n^2}{n+n^2}$

$$\begin{aligned}\lim_{n \rightarrow +\infty} a_n &= \lim_{n \rightarrow +\infty} \frac{2-3n^2}{n+n^2} \\ &= \lim_{n \rightarrow +\infty} \frac{(2-3n^2)/n^2}{(n+n^2)/n^2} \\ &= \lim_{n \rightarrow +\infty} \frac{\frac{2}{n^2} - 3}{\frac{1}{n} + 1} \\ &= -3\end{aligned}$$

$\Rightarrow \{a_n\}$ converges to $L = -3$

(b) $a_n = \frac{4 \sin(n)}{n}$

$$\begin{aligned}-1 &\leq \sin(n) \leq 1 \\ \Rightarrow -\frac{4}{n} &\leq \frac{4 \sin(n)}{n} \leq \frac{4}{n} \\ \Rightarrow 0 &\leq \lim_{n \rightarrow +\infty} \frac{4 \sin(n)}{n} \leq 0 \\ \Rightarrow \lim_{n \rightarrow +\infty} \frac{4 \sin(n)}{n} &= 0\end{aligned}$$

$\Rightarrow \{a_n\}$ converges to $L = 0$

(c) $a_n = (1 - \frac{3}{n})^n$

$$\begin{aligned}\lim_{n \rightarrow +\infty} a_n &= \lim_{n \rightarrow +\infty} (1 - \frac{3}{n})^n \\ &= e^{-3}\end{aligned}$$

$\Rightarrow \{a_n\}$ converges to $L = e^{-3}$

(d) $a_n = \frac{n^2}{e^n}$ (Hint: L'Hopital's rule with $f(x) = \frac{x^2}{e^x}$)

let $f(x) = \frac{x^2}{e^x}$, then $a_n = f(n)$

$$\begin{aligned}\lim_{x \rightarrow +\infty} \frac{x^2}{e^x} &= \lim_{x \rightarrow +\infty} \frac{2x}{e^x} \\ &= \lim_{x \rightarrow +\infty} \frac{2}{e^x} = 0\end{aligned}$$

$$\Rightarrow \lim_{n \rightarrow +\infty} \frac{n^2}{e^n} = 0$$

$\Rightarrow \{a_n\}$ converges to $L = 0$

Problem 2. (4 points) Find the sum of the geometric series $\sum_{n=1}^{+\infty} \frac{(-3)^{n-1}}{4^n}$.

$$a_1 = \frac{(-3)^0}{4^1} = \frac{1}{4}, \quad a_2 = \frac{(-3)^1}{4^2} = -\frac{3}{16}$$

$$\Rightarrow a = a_1 = \frac{1}{4}, \quad r = \frac{a_2}{a_1} = \frac{-3/16}{1/4} = -\frac{3}{4} \Rightarrow \sum_{n=1}^{+\infty} \frac{1}{4} \left(-\frac{3}{4}\right)^{n-1}$$

$$|r| = \frac{3}{4} < 1 \Rightarrow \sum_{n=1}^{+\infty} \frac{(-3)^{n-1}}{4^n} \text{ converges}$$

$$\text{and } \sum_{n=1}^{+\infty} \frac{(-3)^{n-1}}{4^n} = \frac{a}{1-r} = \frac{\frac{1}{4}}{1 - (-\frac{3}{4})} = \frac{\frac{1}{4}}{\frac{7}{4}} = \frac{1}{7}$$